تجمع جازان الصحي
Jazan Health Cluster

STROKE UNIT DELIRIUM SCREENING PROTOCOL

Document ID: SU-006-PROT-01 | Parent Policy: SU-006 — NIHSS Assessment, Neurological Monitoring and Dysphagia/Swallow Screening | Version: 1.0 | Controlled Copy

Certification Alignment Note: Certification framing has been corrected for the American Stroke Association International Stroke Center Certification pathway; delirium remains a local clinical-quality sub-protocol rather than a standalone core stroke performance measure.

Standardized Delirium Recognition, Screening, and Escalation Pathway for Acute Stroke and Neuroscience Unit Patients

Applicable Units: Acute Stroke Unit • ICU / critical care • Comprehensive/Primary Stroke Center Inpatient Beds

Document Type: Stroke Program Clinical Sub-Protocol (Companion to Hospital Delirium Screening Protocol)

Guideline Basis: American Heart Association / American Stroke Association (AHA/ASA) Guidelines for Stroke Care; Get With The Guidelines–Stroke; American Stroke Association International Comprehensive Stroke Center Certification Standards

Prepared for review by: Stroke Program Medical Director • Stroke Coordinator • Nursing Practice Council • Quality & Patient Safety Committee

Effective Date: [Insert Date] Review Cycle: Annual Version: 1.0

Document Owner: [Insert Stroke Coordinator/Title] Approved by: [Insert Name/Title]

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1. Purpose and Scope

1.1 Purpose

This sub-protocol adapts the hospital's general Delirium Screening Protocol for use in patients admitted with acute ischemic stroke, intracerebral hemorrhage, subarachnoid hemorrhage, or transient ischemic attack. Post-stroke delirium (PSD) occurs in an estimated 10–48% of stroke admissions and is independently associated with longer length of stay, higher mortality, and worse functional outcome at discharge. Standard delirium tools were developed and validated primarily in non-neurological populations; this protocol specifies the adjustments required for accurate use in patients with stroke-related neurological deficits.

1.2 Scope

Applies to all patients admitted to the Acute Stroke Unit, ICU / critical care, or any inpatient bed under the stroke service, regardless of stroke subtype (ischemic, hemorrhagic, or TIA under observation).

1.3 Relationship to General Hospital Protocol

This document does not replace the hospital's general Delirium Screening Protocol; it modifies tool selection, timing, and interpretation for the stroke population and should be used alongside it. Where this protocol is silent, the general hospital protocol governs.

2. Guideline and Accreditation Basis

Unlike general hospital accreditation, stroke care carries a certification pathway in which the American Heart Association/American Stroke Association (AHA/ASA) plays a direct role. For this project, certification is pursued through the American Stroke Association International Stroke Center Certification program, but its clinical guidelines and Get With The Guidelines–Stroke (GWTG-Stroke) registry measures form the clinical basis that accrediting bodies build stroke certification survey standards on.

Framework	Relevance to This Protocol
AHA/ASA Guidelines for the Early Management of Acute Ischemic Stroke / Guidelines for Management of Spontaneous ICH	Clinical basis for post-stroke monitoring, neurological deterioration recognition, and complication surveillance, including delirium as a recognized complication.
AHA/ASA Get With The Guidelines–Stroke (GWTG-Stroke)	National registry and quality measure set that stroke centers report against; institutions may elect to track delirium as a locally added quality measure.
American Stroke Association International Stroke Center Certification Primary Stroke Center, Thrombectomy-Capable, and Comprehensive Stroke Center Certification	Requires standardized neurological assessment and complication-monitoring protocols on the stroke unit; this delirium sub-protocol supports that requirement.

Framework	Relevance to This Protocol
KFCH / Jazan Health Cluster nursing reassessment, patient-safety, and restraint policies	Nursing reassessment standards; restraint standards requiring exclusion of a reversible cause (including delirium) before restraint use.

Certification Note

If your stroke center is pursuing or maintaining Joint Commission or DNV stroke certification, route this document through your Stroke Program Medical Director and Stroke Coordinator for inclusion in the certification evidence binder, and cite your specific certification level (Primary, Thrombectomy-Capable, or Comprehensive) and standard number.

3. Why Standard Delirium Tools Require Adaptation in Stroke

Validation studies conducted in stroke units and neuro-ICUs consistently show that standard delirium screening tools perform differently in stroke patients than in general medical populations, for a specific reason: stroke itself can produce inattention, disorientation, aphasia, neglect, and fluctuating arousal that mimic delirium features even when no true delirium is present.

- The Confusion Assessment Method (CAM) remains the best-validated tool against DSM criteria in acute stroke, but its sensitivity is reduced compared with non-stroke populations, particularly in more severe strokes, because stroke-related deficits and delirium are difficult to distinguish.
- The 4AT and Nursing Delirium Screening Scale (NuDESC) show higher sensitivity but lower specificity in stroke — patients with neglect, aphasia, or disorientation from the stroke itself are frequently misclassified as delirium-positive (false positives) by these tools.
- The Intensive Care Delirium Screening Checklist (ICDSC) has shown strong sensitivity and specificity in neuro-ICU and stroke unit cohorts and is a reasonable alternative where CAM-ICU is difficult to apply.
- Impaired consciousness (lower RASS), expressive aphasia, impaired language comprehension, and hemineglect are all independently associated with positive delirium screens regardless of true delirium status — these findings must be interpreted alongside the neurological exam, not in isolation.

Core Principle

No delirium screen should be interpreted without first establishing which cognitive/language findings are explained by the patient's known stroke deficits (per the most recent NIHSS and documented exam) and which represent a new or fluctuating change from that stroke-related baseline. A screen is only meaningfully “positive” when it reflects a change superimposed on the established post-stroke baseline.

4. Stroke-Specific Delirium Risk Factors

In addition to the general hospital high-risk criteria (age 65+, dementia, polypharmacy, sensory impairment, prior delirium), the following stroke-specific factors increase delirium risk and warrant high-risk (every-shift) screening:

- Right hemisphere stroke or strokes involving the insula, temporo-parietal cortex, or thalamus
- Larger infarct volume or extensive white matter disease on imaging
- Hemorrhagic stroke (intracerebral or subarachnoid hemorrhage)
- Higher stroke severity (higher NIHSS score on admission)
- Pre-existing cognitive impairment or prior stroke
- Post-thrombectomy or post-thrombolysis status (first 24–48 hours)
- Concurrent infection (aspiration pneumonia, UTI — common after dysphagia or catheterization)
- Seizure or non-convulsive status epilepticus as a differential diagnosis
- ICU/stroke unit environment: sleep disruption from frequent neuro checks, continuous monitoring, unfamiliar surroundings

5. Screening Workflow

5.1 Establishing the Neurological and Cognitive Baseline

1. On admission, document baseline pre-stroke mentation (from patient/family/prior records) separately from the current post-stroke neurological exam.
2. Document current deficits from the admission NIHSS and neurological exam: presence and type of aphasia (expressive/receptive/global), neglect, hemianopia, dysarthria, and level of consciousness. This becomes the stroke-related “reference exam” against which delirium screens are interpreted.
3. Communicate the reference exam findings at each shift handoff so that subsequent screeners do not mistake a stable, known deficit for a new delirium feature.

5.2 Screening Tool Selection

Clinical Situation	Preferred Tool
Stroke unit, patient verbal and able to participate	CAM (preferred — best validated against DSM criteria in stroke); interpret alongside reference exam
Stroke unit or neuro-ICU, significant aphasia or reduced arousal limiting CAM use	ICDSC (strong sensitivity/specificity in neuro-ICU/stroke populations; relies more on observed behavior than verbal responses)
Mechanically ventilated / post-thrombectomy sedated patient	CAM-ICU, only once RASS is −3 or higher; if persistently non-assessable, use ICDSC observational items

Clinical Situation	Preferred Tool
Rapid general screen, e.g., prior to transfer out of stroke unit	4AT — interpret positive results cautiously in patients with known aphasia/neglect (higher false-positive rate); confirm with CAM or clinical judgment before escalating

5.3 Timing

Setting	Minimum Frequency
ICU / critical care / first 48 hours post-thrombectomy or post-tPA	Every shift (every 12 hours), aligned with scheduled neuro checks
Acute stroke unit — high-risk criteria present (Section 4)	Every shift (every 12 hours)
Acute stroke unit — standard risk	Once every 24 hours
Any setting — acute change in behavior, new agitation, or new unexplained drop in arousal	Immediate ad hoc screening, and immediate neurological reassessment to exclude recurrent stroke, hemorrhagic transformation, or seizure

Where feasible, align delirium screening with scheduled neuro-check intervals so screening does not add additional sleep disruption beyond what is clinically necessary.

6. Differential Diagnosis of a Positive Screen in Stroke Patients

Before attributing a positive delirium screen to delirium alone, the stroke team must exclude the following stroke-specific causes of acute mental status change, several of which require urgent imaging or EEG rather than delirium management alone:

Differential	Initial Work-Up
Hemorrhagic transformation or hematoma expansion	Urgent non-contrast head CT
Recurrent or progressing stroke	Urgent non-contrast head CT ± CTA/MRI, repeat NIHSS
Non-convulsive seizure / status epilepticus	STAT EEG if clinical suspicion, especially with fluctuating unexplained unresponsiveness
Increased intracranial pressure / hydrocephalus (large infarct, hemorrhage, or SAH)	CT head, neurosurgical consultation as indicated
Infection (aspiration pneumonia, UTI, line infection)	Fever work-up, chest exam/imaging, urinalysis as indicated

Differential	Initial Work-Up
Metabolic derangement (hyponatremia, especially with SAH; hyperglycemia; hypoxia)	Basic metabolic panel, glucose, pulse oximetry/ABG
Medication effect (sedatives, anticholinergics, new antiepileptic load)	Medication reconciliation and review
Alcohol or benzodiazepine withdrawal	Withdrawal risk assessment per hospital protocol

Only after these stroke-specific and general medical causes have been reasonably excluded or are being concurrently treated should the positive screen be managed primarily as delirium.

7. Escalation Pathway

4. Notify the bedside RN and stroke team (resident/APP/attending) of any positive screen or acute change within the same shift; changes suggestive of new neurological deficit require immediate notification and STAT non-contrast head CT per stroke protocol, not delayed to routine delirium work-up.
5. Complete the differential diagnosis work-up in Section 6 concurrently with delirium management — do not assume delirium until stroke-specific causes are addressed.
6. Review all scheduled and PRN medications for deliriogenic agents, with particular attention to benzodiazepines, opioids, and anticholinergics, which may also blunt the neurological exam and mask a new deficit.
7. Initiate the non-pharmacologic prevention bundle (Section 8) for all patients with a positive screen or 2 or more stroke-specific risk factors.
8. Document reference-exam-adjusted interpretation: state explicitly which screen findings are attributable to known stroke deficits versus new/fluctuating findings.
9. Continue high-risk screening frequency until two consecutive screens are negative (or stable relative to the documented reference exam) and the clinical team agrees the acute episode has resolved.
10. Physical or chemical restraint use requires provider order and must follow exclusion of reversible causes; avoid antipsychotics with significant anticholinergic or QT-prolonging properties without pharmacy/cardiology input, particularly in patients on QT-prolonging antiarrhythmics or with electrolyte derangement.

8. Non-Pharmacologic Prevention Bundle (Stroke-Adapted)

- Early mobilization per stroke unit mobility protocol, coordinated with physical therapy and fall-risk precautions
- Swallow screen completed before any oral intake (medications or food) to reduce aspiration-related infection risk, a major delirium precipitant in stroke

- Sensory aids in use (glasses, hearing aids) to reduce disorientation on top of stroke-related deficits
- Communication boards, picture cards, or speech-language pathology support for patients with aphasia to support orientation and reduce misclassification of communication difficulty as confusion
- Cluster neuro checks and vitals where clinically safe to protect sleep, without compromising required neurological monitoring frequency
- Family presence and familiar objects at bedside
- Daily medication review with pharmacy for deliriogenic and neurologically sedating agents
- Visual field and neglect-aware room setup (place call bell, personal items, and staff approach on the patient's unaffected side when neglect is present)

9. Roles and Responsibilities

Role	Responsibility
Bedside RN / Stroke Unit RN	Performs screening at required intervals using the correct tool for the patient's communication ability; documents reference-exam-adjusted interpretation; notifies stroke team of positive screens or acute changes.
Stroke Team (Resident/APP/Attending)	Responds to notification; orders differential work-up (imaging, EEG, labs as indicated); documents whether findings are attributable to stroke deficit, delirium, or a new neurological event.
Speech-Language Pathology	Supports communication assessment in aphasic patients; helps distinguish language deficit from disorganized thinking; completes swallow screening.
Pharmacist	Reviews medications for deliriogenic and neuro-sedating agents; flags interactions with antiepileptics or antithrombotics.
Stroke Coordinator	Maintains this sub-protocol, tracks delirium-related metrics for stroke certification reporting, coordinates staff training.
Quality & Patient Safety Committee	Monitors screening compliance and outcomes; incorporates delirium metrics into stroke program quality reporting as applicable.

10. Documentation Requirements

- Baseline pre-stroke mentation (on admission)
- Reference neurological/cognitive exam (deficits attributable to the stroke itself), updated each shift
- Screening tool used, rationale for tool selection if not the default, and total score/findings
- Explicit statement distinguishing known stroke-related findings from new/fluctuating delirium features
- Differential diagnosis work-up performed and results (imaging, EEG, labs) for any positive screen or acute change
- Notification time and stroke team response

- Prevention bundle elements implemented
- Resolution criteria met (two consecutive screens negative or stable relative to reference exam)

11. Quality Monitoring and Metrics

- Percentage of eligible stroke unit patients screened per required interval (target: 90% or greater)
- Time from positive screen/acute change to stroke team notification and, where indicated, repeat imaging (target: under 60 minutes for acute change)
- Rate of delirium among stroke unit patients, stratified by stroke subtype and NIHSS severity
- Rate of positive screens subsequently attributed to a stroke-specific cause (hemorrhagic transformation, recurrent stroke, seizure) rather than delirium
- Restraint utilization among stroke unit patients with positive delirium screens
- Staff competency completion for stroke-adapted delirium screening

12. Training and Competency

11. All stroke unit and neuro-ICU nursing staff complete competency training on CAM, ICDSC, and the stroke-specific interpretation principles in Section 3 and Section 6, in addition to general hospital delirium screening competency.
12. Training explicitly covers distinguishing aphasia, neglect, and disorientation attributable to stroke from true inattention and disorganized thinking of delirium, using case-based examples.
13. Competency is validated by direct observation from the Stroke Coordinator or unit educator and documented in the staff competency record; annual refresher required.
14. New hires to the stroke unit complete this competency during unit-based orientation prior to independent patient assignment.

13. Review, Revision, and Approval

This sub-protocol is reviewed annually alongside the general hospital Delirium Screening Protocol, or sooner if triggered by updated AHA/ASA stroke guidelines, a change in stroke center certification requirements, or a relevant sentinel event. Revisions require review and approval by the Stroke Program Medical Director, Stroke Coordinator, Nursing Practice Council, and Quality & Patient Safety Committee.

Version	Date	Summary of Change	Approved By
1.0	[Insert Date]	Initial release	[Insert Name/Title]

14. References

1. Shaw RC, Walker G, Elliott E, Quinn TJ. Occurrence Rate of Delirium in Acute Stroke Settings: Systematic Review and Meta-Analysis. Stroke.

2. Lees R, Corbet S, Johnston C, et al. Test Accuracy of Short Screening Tests for Diagnosis of Delirium or Cognitive Impairment in an Acute Stroke Unit Setting. Stroke.

3. Rizk A, et al. Screening for Delirium in Acute Stroke. Stroke (AHA/ASA journal).

4. Kotfis K, et al. Delirium Screening in Neurocritical Care and Stroke Unit Patients: Influence of Neurological Deficits on CAM-ICU and ICDSC Outcome.

5. Wilson JE, et al. Course and Recognition of Poststroke Delirium. Stroke.

6. American Heart Association/American Stroke Association. Guidelines for the Early Management of Patients With Acute Ischemic Stroke.

7. American Heart Association/American Stroke Association. Get With The Guidelines–Stroke Program Overview.

Before Finalizing

Insert your stroke center's name, certification level (Primary/Thrombectomy-Capable/Comprehensive), policy number, EHR flowsheet names, and committee approval signatures before distribution. Confirm tool selection (CAM vs. ICDSC vs. CAM-ICU vs. 4AT) with your Stroke Program Medical Director to match your unit's patient acuity and staffing model.